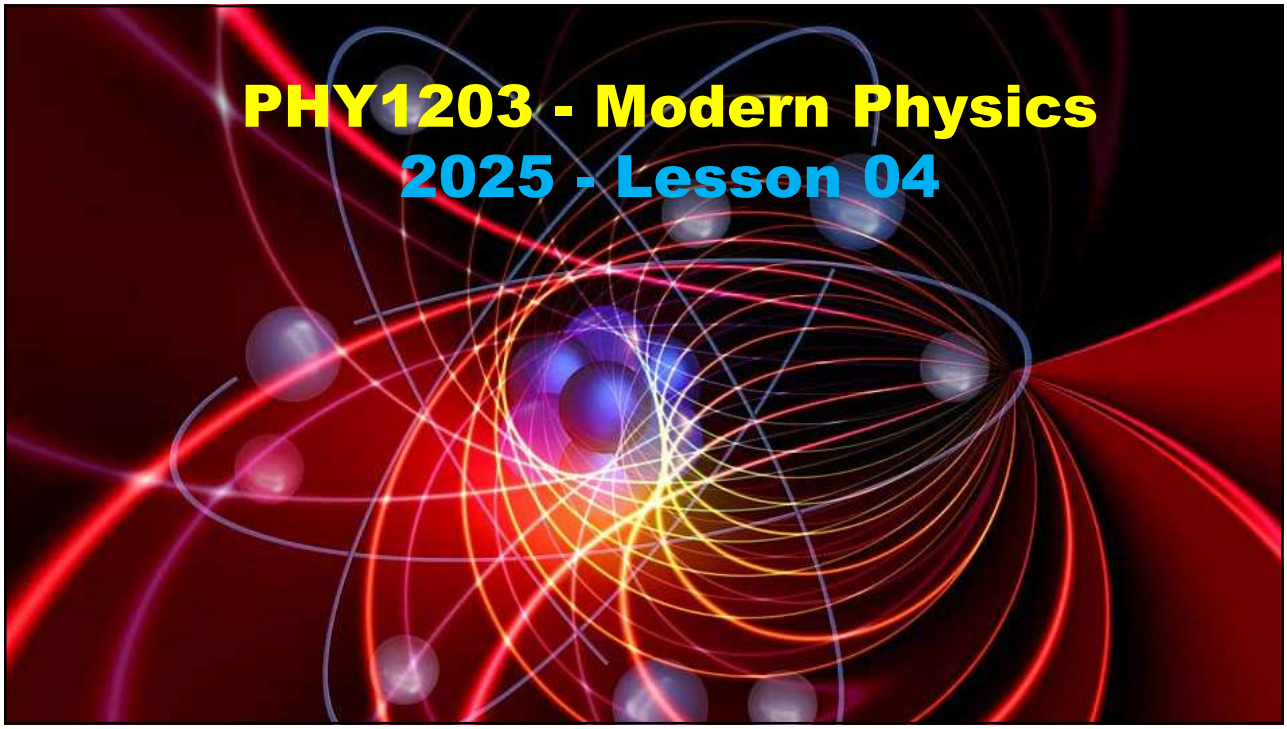


Uncertainty Principle

The principle was formulated by Werner Heisenberg in 1927.

This is a fundamental concept in quantum mechanics.

It gives a fundamental limit to the precision of a particle that can be simultaneously known.



Werner Heisenberg
(1901-1976)

Heisenberg's Uncertainty Principle (1927)

It is physically impossible to simultaneously measure the exact position and the exact momentum precisely (or accurately).

$$\Delta x . \Delta p \geq \frac{\hbar}{2}$$

where Δx = uncertainty in position

Δp = uncertainty in momentum

$\hbar = \frac{h}{2\pi}$ = reduced Planck's constant $(= 1.06\times10^{-34}$ J s)

- If you try to measure the position of a particle very accurately (very small Δx), you will introduce a large uncertainty in its momentum (large Δp).
- Conversely, if you measure the momentum precisely, the position becomes highly uncertain.

Example (1)

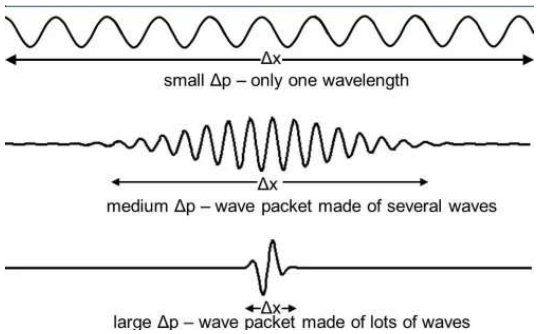
Consider an electron in an atom,

the position uncertainty of $\Delta x \approx 10^{-10}$ m .

That leads to momentum uncertainty of $\Delta p \geq \frac{\hbar}{2\Delta x}$

$$\Delta p \geq 1.06\times10^{-24} \text{ kg m s}^{-1}$$

Example (2)



Structure of the atom

- Dalton's hard sphere model (1803)
- Thompson's “Plum pudding” model (1904)
- Rutherford Model (1911)
- Bohr Model (1913)
- Quantum Mechanical View

Dalton's atomic model



John Dalton

(1766 - 1844)

John Dalton's hard sphere atomic model was developed around 1803.

He proposed that atoms are tiny, indivisible, and indestructible spheres.

This model was a foundation of early atomic theory and helped explain the behavior of matter in chemical reactions.

Main features of Dalton's model

- **Atoms are indivisible.**
  Dalton believed atoms were the smallest units of matter and could not be broken down further.
- **Atoms are solid and hard.**
  The model described atoms as solid, impenetrable spheres, like tiny billiard balls.

- **Atoms of a given element are identical.**
  Atoms of the same element were thought to have the same mass and properties.
- **Atoms of different elements are different.**
  Different elements were distinguished by having different masses and properties for their atoms.
- **Atoms combine to form compounds.**
  Atoms of different elements could combine in simple whole-number ratios to form chemical compounds.
- **Atoms rearrange in chemical reactions.**
  Chemical reactions were seen as the rearrangement of atoms, not the creation or destruction of atoms.

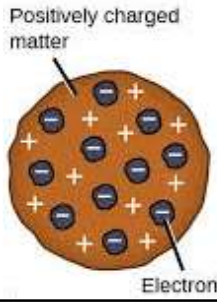
Dalton's ideas provided a scientific basis for understanding the composition and behavior of matter. His model was based on experimental observations of his work with gases and pressure.

Thompson's "Plum pudding" Model

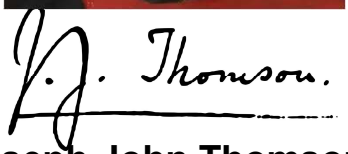
J.J. Thomson proposed the plum pudding model of the atom in 1904 following his discovery of the **electron** in 1897.

It described the atom as a sphere of positive charge (pudding) with negatively charged electrons (plum), like plums in a pudding.









Joseph John Thomson (1856-1940)

The model explained the neutral atoms. The positive and negative charges are equal in magnitude canceling each other out.

- **Positive Sphere:**
  The model proposed that the atom is primarily composed of a sphere of positive charge.
- **Embedded Electrons:**
  Scattered throughout this sphere are negatively charged electrons, similar to plums in a traditional plum pudding.
- **Neutral Atom:**
  The overall charge of the atom is neutral because the positive charge of the sphere is equal in magnitude to the negative charge of the electrons.

This model was proposed after the discovery of the electron and before the discovery of the nucleus.

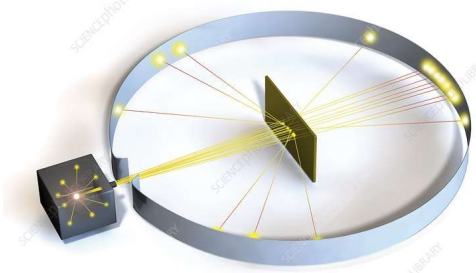
But this model could not explain

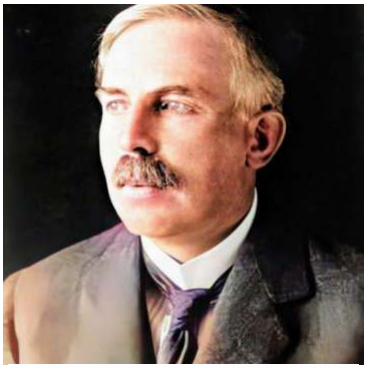
- the stability of the atom,
- the nucleus' position or
- the results of Rutherford's gold foil experiment.

Rutherford Model

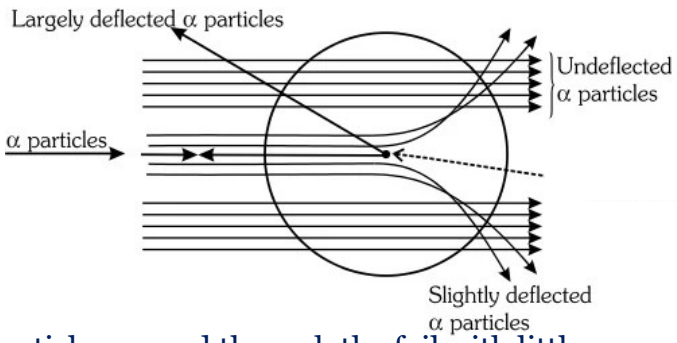
Ernest Rutherford performed alpha particle scattering experiment. He directed a beam of α-particles at a thin gold foil to investigate the atomic structure and the nucleus in 1911.

A zinc sulfide screen was used to detect the scattered α-particles.





Ernest Rutherford
(1871-1937)

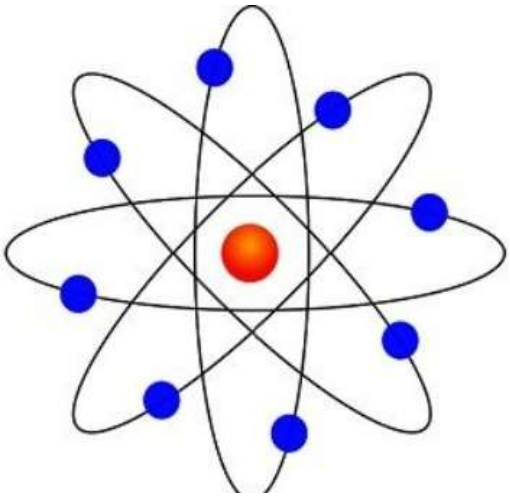


- Most α-particles passed through the foil with little or no deflection,

  $\Rightarrow$ indicating that atoms are mostly empty space.

- A small fraction of α-particles were deflected at large angles, very few were bounced back.

  $\Rightarrow$ He suggested that a small, dense, positively charged region within the atom and it was called the nucleus which must be very small compared to the overall size of the atom.



Bohr Model

Niels Bohr proposed the Bohr atomic model in 1913.

He described the atom as a small, positively charged nucleus surrounded by electrons in specific, quantized orbits.

These orbits are at fixed distances from the nucleus, and electrons can only move in these defined paths, not in between.



Niels Bohr (1885-1962)

Summery of the Bohr model

- **Quantized Energy Levels**

  Electrons orbit the nucleus in well defined paths called orbits or shells, and each orbit has a fixed energy level.

- **No Radiation in Orbits**

  Electrons moving in these orbits do not emit radiation and therefore, maintain a stable energy level.

- **Quantum Jumps**

  Electrons can jump to higher energy levels (away from the nucleus) by absorbing energy and to lower energy levels by emitting energy.

- **Angular Momentum Quantization**

  The angular momentum of an electron in an orbit is quantized, meaning it can only take on specific discrete values.

- **Explanation of Atomic Spectra**

  The Bohr model was successful in explaining the line spectrum of hydrogen. When an electron transitions between energy levels, it emits or absorbs a photon of a specific energy and frequency, corresponding to a specific line in the spectrum.

The Bohr model has limitations because it only works for hydrogen but fails to accurately predict the spectra of atoms with many electrons.